

Data Model, Query and State Walkthrough

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Team Name: SEG26-33 Meraki Arts

Project Title: Meraki Arts Africa Website

Individual GitHub Repository:

<https://github.com/Elorm07/dcit208-portfolio.git>

Team GitHub Repository: <https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-seg26-33-merakiartsafrica.git>

Related github issue url: <https://github.com/Elorm07/dcit208-portfolio/issues/1#issue-5050435321>

A. Project and Feature Context

- i. Selected Feature= Event participant registers for an event
- ii. Feature Description= The Event Registration feature allows visitors to register for upcoming events organized by Meraki Arts Africa. Users provide their personal information, and the system stores their registration while ensuring that the event exists and still has available spaces.
- iii. Related User Story= US-04: Event Registration. As a visitor, I want to register for an upcoming event so that I can participate in Meraki Arts Africa activities.
- iv. Related GitHub Issue= Paste the GitHub issue link for the Event Registration feature.

B. Entities and Attributes

The Event Registration feature requires three main entities: Event, Participant, and Registration.

1. Event Entity= This entity stores information about events organized by Meraki Arts Africa.

Attribute	Data Type	Description
eventId (PK)	UUID	Unique identifier for each event
title	String	Name of the event
description	Text	Brief description of the event
eventDate	Date	Date the event will take place
startTime	Time	Event start time
location	String	Venue of the event

Primary Key: eventId

2. Participant Entity= This entity stores information about people registering for events.

Attribute	Data Type	Description
participantId (PK)	UUID	Unique identifier for each participant
firstName	String	Participant's first name
lastName	String	Participant's last name
email	String	Participant's email address
phoneNumber	String	Participant's contact number

Primary Key: participantId

3. Registration Entity= This entity records each registration made by a participant for an event.

Attribute	Data Type	Description
registrationId (PK)	UUID	Unique identifier for each registration
participantId (FK)	UUID	References the participant
eventId (FK)	UUID	References the event
registrationDate	DateTime	Date and time of registration
registrationStatus	Enum (Pending, Confirmed, Cancelled)	Current registration status

Primary Key: registrationId

Foreign Keys:

participantId references Participant.participantId

eventId references Event.eventId

C. Relationships and Cardinality

The Event Registration feature uses three entities: Participant, Event, and Registration. The Registration entity acts as a link between participants and events.

- **Relationship 1: Participant → Registration**

Relationship: One-to-Many (1:M)

One participant can register for multiple events.

Each registration belongs to only one participant.

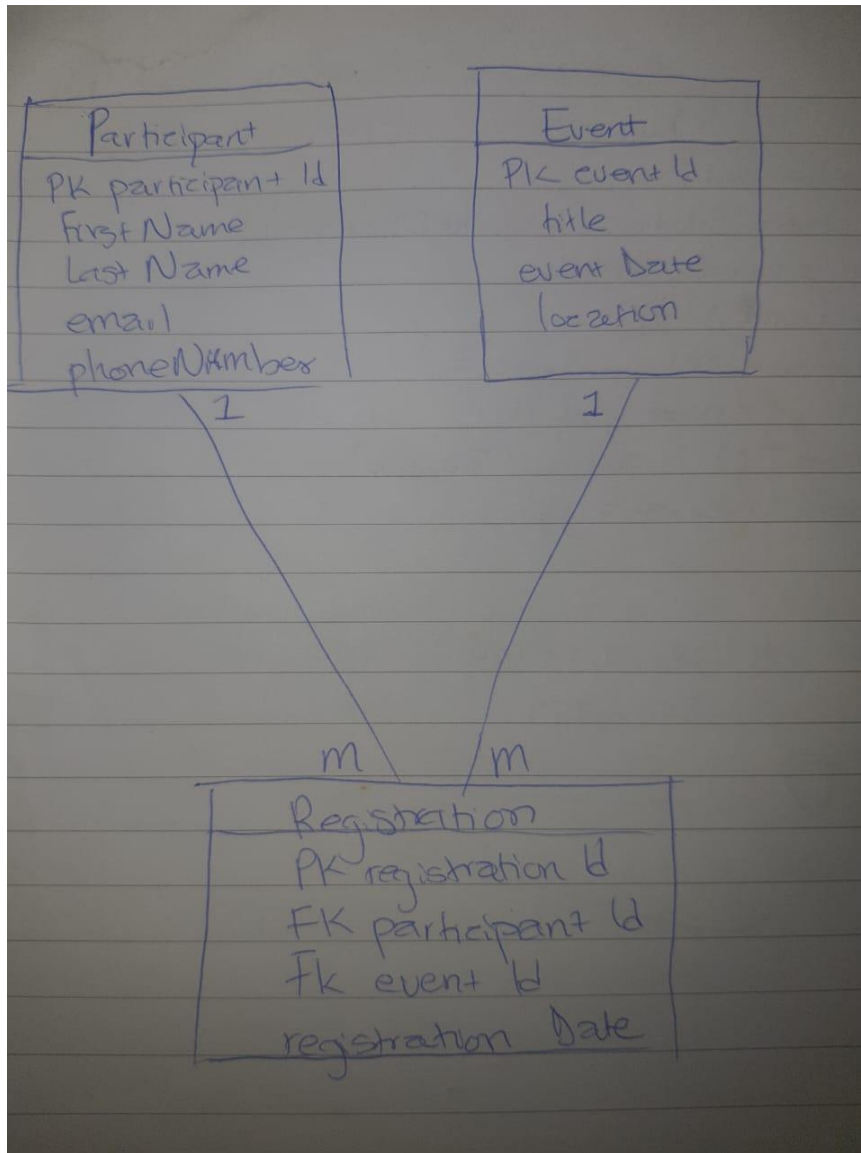
- **Relationship 2: Event → Registration**

Relationship: One-to-Many (1:M)

One event can have many participants registered.

Each registration belongs to only one event.

A simple hand-drawn ERD or table relationship sketch



D. Validation and Constraints

To ensure that the Event Registration feature functions correctly and maintains data integrity, the system should validate user input and enforce business rules at the frontend, backend, and database levels.

Validation Rule / Constraint	Description	Layer
First name is required	The participant must provide a first name before submitting the form.	Frontend + Backend
Last name is required	The participant must provide a last name.	Frontend + Backend
Email is required and must be valid	The email address must follow a valid format (e.g., user@example.com).	Frontend + Backend
Phone number is required	A valid phone number must be provided.	Frontend + Backend
Event ID must exist	The selected event must exist in the database before registration is saved.	Backend + Database
Registration ID must be unique	Each registration record must have a unique identifier.	Database
Participant email should not be registered twice for the same event	A participant cannot register multiple times for the same event.	Backend + Database
Event capacity must not be exceeded	Registration is only allowed if the number of registered participants is below the event capacity.	Backend
Registration status must be valid	The status can only be Pending, Confirmed, or Cancelled.	Backend + Database
Foreign key constraints	participantId must reference an existing participant, and eventId must reference an existing event.	Database

Business Rules

1. A participant can register for multiple events.
2. An event can have multiple participants.
3. A participant cannot register more than once for the same event.
4. Registration is not allowed if the event has reached its maximum capacity.
5. Only administrators are allowed to update or cancel registration records after they have been created.

E. Synthetic Sample Data

The following records are synthetic sample data created for demonstration purposes only. They do not represent real client or user information.

Entity 1: Event

Event ID	Title	Event Date	Location	Capacity	Status
EVT001	Art Exhibition 2026	20 Aug 2026	Accra International Conference Centre	100	Upcoming
EVT002	Creative Skills Workshop	10 Sept 2026	Meraki Arts Centre, Accra	50	Upcoming

Entity 2: Participant

Participant ID	First Name	Last Name	Email	Phone Number
PAR001	Ama	Mensah	ama.mensah@example.com	0241234567
PAR002	Kojo	Owusu	kojo.owusu@example.com	0559876543

Entity 3: Registration

Registration ID	Participant ID	Event ID	Registration Date	Registration Status
REG001	PAR001	EVT001	02 Aug 2026 10:15 AM	Confirmed
REG002	PAR002	EVT002	02 Aug 2026 2:45 PM	Pending

F. Query or State Walkthrough

Selected State Change: Event Registration

The selected state change is when a visitor submits the event registration form to register for a Meraki Arts Africa event.

Input

The participant enters the following information:

- First name
- Last name
- Email address
- Phone number
- Selected event

The system also receives the eventId of the chosen event.

Data Touched

The following entities are accessed:

- Event – to verify that the event exists and has available capacity.
- Participant – to check whether the participant already exists or to create a new participant record.
- Registration – to create a new registration linking the participant to the selected event.

Rules Applied

Before saving the registration, the system checks that:

- All required fields have been completed.
- The email address is in a valid format.
- The selected event exists.
- The event has not reached its maximum capacity.
- The participant has not already registered for the same event.

Storage Process

If all validation checks pass:

1. The participant information is saved (or an existing participant record is used).
2. A new registration record is created.
3. The registration is linked to both the participant and the selected event using their IDs.
4. The registration status is stored as Confirmed (or Pending, depending on the system design).

System Response

If the registration is successful:

- The registration is saved in the database.
- A success message confirming the registration is displayed to the participant.

If the registration is unsuccessful:

- No registration is stored.
- The participant receives an error message explaining the reason, such as:
 1. "This event is full."
 2. "You have already registered for this event."
 3. "Please complete all required fields."

G. Privacy and Access Control

The Event Registration feature collects personal information from participants. The system must protect this information and ensure that only authorized users can access or modify it.

Sensitive Data

The following data items are considered sensitive:

- Participant's full name
- Email address
- Phone number

Access Control

User Role	View Registration Details	Create Registration	Edit Registration	Delete Registration
Visitor/Participant	Only their own registration	Yes	No	No
Administrator	Yes	Yes	Yes	Yes

- Participants can register for an event and view only their own registration details.
- Administrators can view, update, or remove registrations when necessary.
- Unauthorized users should not be able to access another participant's personal information.

Information That Should Not Be Pasted into AI Tools are

- Real participant names, email addresses, or phone numbers
- Client databases or exported registration records
- Login credentials and passwords

H. Failure Scenarios

Failure Scenario 1= Duplicate Registration

Scenario= A participant attempts to register for the same event more than once using the same email address.

What the System Should Do

- Check whether a registration already exists for the participant and the selected event.
- Reject the duplicate registration.
- Display an error message such as: "You have already registered for this event."
- Do not create another registration record in the database.

Failure Scenario 2= Event Has Reached Maximum Capacity

Scenario= A participant tries to register for an event that has already reached its maximum number of participants.

What the System Should Do

- Check the number of confirmed registrations against the event's capacity.
- Prevent any additional registrations once the capacity limit has been reached.
- Display a message such as: "Registration is closed because the event is full."
- Ensure that no new registration record is saved.

I. Test Ideas

Test Case	Expected Result
1. Register with valid participant details and an available event.	Registration is successful, and a new registration record is created.
2. Submit the registration form with one or more required fields left blank.	Registration is rejected, and the user is prompted to complete the missing fields.
3. Enter an invalid email address format (e.g., ama@gmail).	Registration is rejected with an error indicating that the email format is invalid.
4. Attempt to register for the same event twice using the same email address.	The system prevents the duplicate registration and displays an appropriate error message.
5. Attempt to register for an event that has already reached its maximum capacity.	Registration is rejected, and the user is informed that the event is full.
6. Attempt to access another participant's registration details without administrator privileges.	Access is denied, and the system protects the participant's personal information.

J. Oral defense video= <https://youtu.be/G2bncbkDA>

K. AI Disclosure

AI was used: Yes.

AI was used to help analyze and interpret the assignment questions, clarify the requirements of each section, and improve the organization and wording of the report. I personally verified all project-specific information, explanations, and responses before including them in this submission. AI did not invent any project data, client evidence, GitHub links, or personal explanations.